

Total pts 50
4 Questions

Question 1 [13 pts]

Midterm Solution
Fall 2025

You are visiting an office building for a job interview. The building has 250 workers. To reduce the health effects of sitting all week, some employees play sports on weekends. Exactly 105 play cricket, 50 play football, 25 play table tennis, and the remaining employees play no sport. Assume these categories are mutually exclusive (no one plays more than one sport).

Among the cricket players, 30 have hypertension (high blood pressure). Among the football players, 8 have hypertension. Among the table tennis players, 12 have hypertension. Among the employees who play no sport, 60 have hypertension.

During your visit, you randomly stop a worker in the building and ask them whether they suffer from hypertension. If they answer in the positive, what is the probability they don't play any sport?

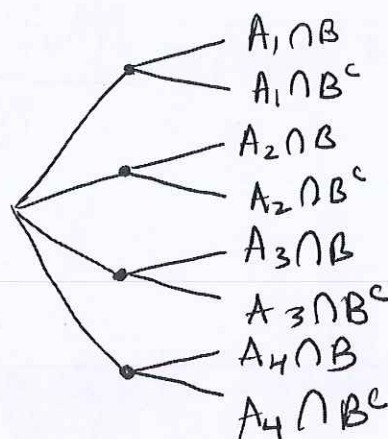
$A_1 = \{ \text{Worker plays cricket} \}$

$A_2 = \{ \text{Worker plays football} \}$

$A_3 = \{ \text{Worker plays table-tennis} \}$

$A_4 = \{ \text{Worker plays no sport} \}$

$B = \{ \text{Worker has hypertension} \}$



Using Bayes' Rule:

$$P(A_4|B) = \frac{P(A_4) \cdot P(B|A_4)}{P(B)} \quad \text{--- (1) [2]}$$

$$P(A_1) = \frac{105}{250}, \quad P(A_2) = \frac{50}{250}, \quad P(A_3) = \frac{25}{250}, \quad P(A_4) = \frac{70}{250}$$

$$P(B|A_1) = \frac{30}{105}, \quad P(B|A_2) = \frac{8}{50}, \quad P(B|A_3) = \frac{12}{25}, \quad P(B|A_4) = \frac{60}{70}$$

$$P(B^c|A_1) = \frac{75}{105}, \quad P(B^c|A_2) = \frac{42}{50}, \quad P(B^c|A_3) = \frac{13}{25}, \quad P(B^c|A_4) = \frac{10}{70}$$

Explicitly or Implied [4]

By Total Probability Theorem:-

$$\begin{aligned}P(B) &= P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B) + P(A_4 \cap B) \\&= P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2) + P(A_3) \cdot P(B|A_3) + P(A_4) \cdot P(B|A_4) \\&= \frac{105}{250} \cdot \frac{30}{105} + \frac{50}{250} \cdot \frac{8}{50} + \frac{25}{250} \cdot \frac{12}{25} + \frac{70}{250} \cdot \frac{60}{70} \\&= \frac{110}{250} \quad [4]\end{aligned}$$

From (1)

$$\begin{aligned}P(A_4|B) &= \frac{P(A_4) \cdot P(B|A_4)}{P(B)} = \frac{\frac{70}{250} \cdot \frac{60}{70}}{\frac{110}{250}} = \frac{60}{110} = \frac{6}{11} \\&= 0.545 \quad [3]\end{aligned}$$

For solutions skipping intermediate steps, full credit if correct, little/no partial credit if incorrect.

A Simple Alternative

$$\begin{aligned}P(A_4|B) &= \frac{P(A_4 \cap B)}{P(B)} \\&= \frac{60/250}{\frac{110}{250}} = \frac{60}{110} = \frac{6}{11} \\&= 0.545\end{aligned}$$

Question No 2 [10 pts]

Consider an experiment consisting of two consecutive rolls of a fair tetrahedral die (faces labeled 1,2,3,4). The events A , B and C are defined as:

A : The first roll is even.

B : The second roll is even.

C : The sum of the two rolls is even.

(a) Test whether the events A , B , and C are independent. [4 pts]

(b) Test whether the events A and B are independent. [2 pts]

(c) Test whether the events A and B are independent when conditioned over C . [4 pts]

$$\Omega = \{(1,1), (1,2), (1,3), (1,4), \\ (2,1), (2,2), (2,3), (2,4), \\ (3,1), (3,2), (3,3), (3,4), \\ (4,1), (4,2), (4,3), (4,4)\}$$

$$A = \{(2,1), \underline{(2,2)}, (2,3), \underline{(2,4)}, \\ (4,1), \underline{(4,2)}, (4,3), \underline{(4,4)}\}$$

$$B = \{(1,2), \underline{(2,2)}, (3,2), \underline{(4,2)}, \\ (1,4), \underline{(2,4)}, (3,4), \underline{(4,4)}\}$$

$$C = \{(1,1), (1,3), \underline{(2,2)}, \underline{(2,4)}, \\ (3,1), (3,3), \underline{(4,2)}, \underline{(4,4)}\}$$

(a)

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{2}, P(C) = \frac{1}{2}$$

$$P(A \cap B \cap C) = \frac{1}{4}$$

$$P(A \cap B) = \frac{4}{16} = \frac{1}{4}$$

$$P(A \cap C) = \frac{4}{16} = \frac{1}{4}$$

$$P(B \cap C) = \frac{4}{16} = \frac{1}{4}$$

$$P(A \cap B) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(A) \cdot P(B) \quad \checkmark$$

Pairwise independent

$$P(A \cap C) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(A) \cdot P(C) \quad \checkmark$$

$$P(B \cap C) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(B) \cdot P(C) \quad \checkmark$$

$$P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = P(A) \cdot P(B) \cdot P(C) \quad \times \text{ Not Mutually Independent. [4 pts]}$$

A, B, C are NOT Independent

(b) Since $P(A \cap B) = P(A) \cdot P(B)$,

A & B are Independent

— [2 pts]

(c) $C = \{(1,1), (1,3), (2,2), (2,4), (3,1), (3,3), (4,2), (4,4)\}$

$A|C = \{(2,2), (2,4), (4,2), (4,4)\}$

$$P(A|C) = \frac{4}{8} = \frac{1}{2} \quad [1 \text{ pt}]$$

$B|C = \{(2,2), (2,4), (4,2), (4,4)\}$

$$P(B|C) = \frac{4}{8} = \frac{1}{2} \quad [1 \text{ pt}]$$

$A \cap B|C = \{(2,2), (2,4), (4,2), (4,4)\}$

$$P(A \cap B|C) = \frac{4}{8} = \frac{1}{2}$$

$$P(A \cap B|C) = \frac{1}{2} \neq \frac{1}{2} \cdot \frac{1}{2} = P(A|C) \cdot P(B|C)$$

— [2 pts]

A and B are NOT Independent when Conditioned over C

3. Counting Problem [12 points]

PIN (Personal Identification Number) from a bank is a secret, numeric code that acts as your electronic signature to authenticate your identity during financial transactions, such as ATM withdrawals or point-of-sale purchases. PIN, is usually 4 to 6 digits long. The bank's **current PIN selection policy** is as follows: -

- Length of 4 to 6 digits, i.e., user may select a 4 or a 5 or a 6-digit long PIN.
- May contain numbers (0-9)
- No spaces

a) [4 points] Under the current PIN selection policy, how many possible PINs (N_1) are there?

Due to recent compromises and hacking attempts, a bank is considering that PIN selection should be made more robust against hacking and decoding by intelligent code-breaking machines. The Bank's cyber security guru has proposed the following criteria for the **revised PIN selection policy**: -

- Length of 4 to 6 characters, i.e., user may select a 4 or a 5 or a 6 characters long PIN.
- Includes at least one special character out of the 5 allowed characters, *, #, @, &, !
- No spaces
- May contain numbers (0-9), lower- and upper-case letters (a-z, A-Z)
- Is case-sensitive.

b) [8 points] Under the revised PIN selection, how many possible PINs (N_2) are there?

(a) for a 4-digit Pin

$$\underset{1}{10} \cdot \underset{2}{10} \cdot \underset{3}{10} \cdot \underset{4}{10} = 10^4$$

(1)

for a 5-digit Pin

$$\underset{1}{10} \cdot \underset{2}{10} \cdot \underset{3}{10} \cdot \underset{4}{10} \cdot \underset{5}{10} = 10^5$$

(1)

for a 6-digit pin

$$\underset{1}{10} \cdot \underset{2}{10} \cdot \underset{3}{10} \cdot \underset{4}{10} \cdot \underset{5}{10} \cdot \underset{6}{10} = 10^6$$

(1)

$$\text{Total PINs} = 10^4 + 10^5 + 10^6$$

$$N_1 = 1,110,000$$

(1)

Pg 5

(b) Possible Choices

(12 pts)

$$10 + 26 + 26 + 5$$

Numbers small
 alphabet Capital
 alphabet Special
 Character

= 67 choices

(2 pts)

For a 4 digit PIN

$$= 67 \cdot 67 \cdot 67 \cdot 67 = 67^4 \text{ --- (A)}$$

(A) all possible permutations
let find without 5 special characters

$$= 62 \cdot 62 \cdot 62 \cdot 62 = 62^4 \text{ --- (B)}$$

(B) all permutations but No Special
Character

$$(A) - (B) = 67^4 - 62^4 \text{ --- (C)}$$

(3 pts)

(C) $\rightarrow$ At least 1 Special Character

So Total PINs

$$N_2 = (67^4 - 62^4) + (67^5 - 62^5) + (67^6 - 62^6)$$

(3 pts)

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4. [15 points] Discrete Random Variable

A university IT helpdesk receives support requests each hour. The random variable X (can take on 5 values that represents the number of requests received in an hour, with the following probabilities:

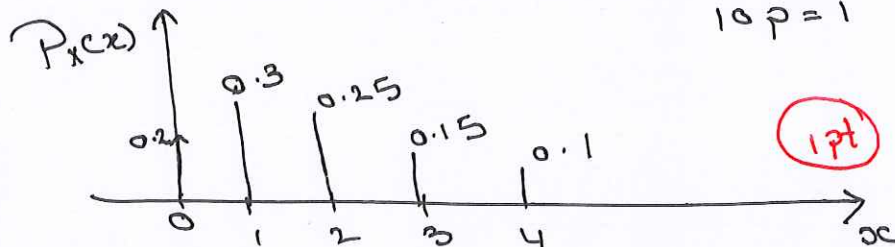
$$P(X=0) = 2p, P(X=1) = 3p, P(X=2) = 2.5p, P(X=3) = 1.5p, P(X=4) = p$$

- (a) [3 points] Calculate p and plot the probability mass function for the random variable X .

The helpdesk has **2 staff members**, and each can handle at most **1 request per hour**. If more than 2 requests arrive, the extra requests are left **unresolved**. Define a new random variable Y = number of unresolved requests in an hour.

- (b) [4 points] Write down and plot the probability mass function for random variable Y .
 (c) [4 points] What is the probability that all requests in an hour cannot be resolved?
 (d) [4 points] Compute $E[Y]$, the expected number of unresolved requests per hour.

(a) For a valid distribution, $2p + 3p + 2.5p + 1.5p + p = 1$
 $10p = 1 \Rightarrow p = 0.1$ (2 pts)

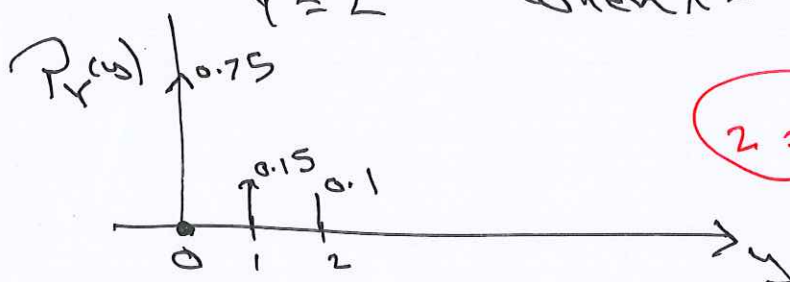


(b) Possible values for RV $Y = \{0, 1, 2\}$ (1 pt)

$Y=0$ when $X=0, 1, 2 \Rightarrow P(Y=0) = 0.75$

$Y=1$ when $X=3 \Rightarrow P(Y=1) = 0.15$

$Y=2$ when $X=4 \Rightarrow P(Y=2) = 0.1$ (1 pt)



(c) Probability that all requests cannot be resolved
 $\Rightarrow Y \geq 1 \Rightarrow P(Y=1 \text{ or } Y=2)$ (2 pts)
 $= P(Y=1) + P(Y=2) = 0.15 + 0.1 = 0.25$ (2 pts)

(d) $E[Y] = \sum_y y P_Y(y)$ (1 pt)
 $= 0 \times 0.75 + 1 \times 0.15 + 2 \times 0.1$ (1 pt)
 $= 0.15 + 0.2 = 0.35$ (1 pt)



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